Answers learning task 10

2026 – 2027

Exercise 7.3: PVC pipe is manufactured with a mean diameter of $2.565\,\text{cm}$ and a standard deviation of $0.008\,\text{cm}$. Find the probability that a random sample of $n = 5$ sections of pipe will have a sample mean diameter greater than $2.560\,\text{cm}$ and less than $2.570\,\text{cm}$.

Answer

Central Limit Theorem: $\bar{X} \sim N(\mu = 2.565, \sigma = 0.008/\sqrt{5} \approx 0.0036)$
0.8377

Exercise 7.7: The compressive strength of concrete (in kN m^{-2}) is normally distributed with $\mu = 17237$ and $\sigma = 345$. Find the probability that a random sample of $n = 5$ specimens will have a sample mean compressive strength that falls in the interval $17\,230\,\text{kN m}^{-2}$ to $17\,305\,\text{kN m}^{-2}$.

Answer

Central Limit Theorem: $\bar{X} \sim N(\mu = 17237, \sigma = 345/\sqrt{5} \approx 154.2887)$
0.1884

Exercise 7.10: Suppose that the random variable X has the continuous uniform distribution

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Suppose that a random sample of $n = 27$ observations is selected from this distribution. What is the approximate probability distribution of $\bar{X} + 1/2$? Find the mean and variance of this quantity.

Answer

Central Limit Theorem: $\bar{X} + 1/2 \sim N\left(\mu = 1, \sigma = 1/18\right)$. The mean is equal to 1 and the variance equal to $\left(\frac{1}{18}\right)^2 = \frac{1}{324}$.

Exercise 7.11: Suppose that X has a discrete uniform distribution

$$f(x) = \begin{cases} 1/4, & x = 1, 2, 3, 4 \\ 0, & \text{otherwise} \end{cases}$$

A random sample of $n = 36$ is selected from this population. Find the probability that the sample mean is greater than 2.3 but less than 2.7, assuming that the sample mean would be measured to the nearest tenth.

Answer

Central Limit Theorem: $\bar{X} \sim N(\mu = 2.5, \sigma = \frac{1}{12}\sqrt{5})$
Correction for continuity, without rounding to one decimal: $P(2.3 < \bar{X} < 2.7) = 0.7169$
Correction for continuity, rounded to one decimal (notice that we need to have unrounded values between 2.35 and 2.65, to get rounded values of 2.4, 2.5 or 2.6): $P(2.35 < \bar{X} < 2.65) = 0.5792$

Exercise 7.12: The amount of time that a customer spends waiting at an airport check-in counter is a random variable with mean $8.2\,\text{min}$ and standard deviation $6\,\text{min}$. Suppose that a random sample of $n = 36$ customers is observed. Find the probability that the average time waiting in line for these customers is

(a) Less than $10\,\text{min}$
(b) Between $5\,\text{min}$ and $10\,\text{min}$
(c) Less than $6\,\text{min}$

Answer

Central Limit Theorem: $\bar{X} \sim N(\mu = 8.2, \sigma = 6/\sqrt{36} = 1)$

(a) 0.9641 (b) 0.9634 (c) 0.0139

Exercise 7.13: A random sample of size $n_1 = 16$ is selected from a normal population with a mean of 75 and a standard deviation of 8. A second random sample of size $n_2 = 9$ is taken from another normal population with mean 70 and standard deviation 12. Let $\bar{X}_1$ and $\bar{X}_2$ be the two sample means. Find:

(a) The probability that $\bar{X}_1 - \bar{X}_2$ exceeds 4

(b) The probability that $2.5 \leq \bar{X}_1 - \bar{X}_2 \leq 5.5$

Answer

Central Limit Theorem: $\bar{X}_1 \sim N(\mu = 75, \sigma = 8/\sqrt{16} = 2)$ and $\bar{X}_2 \sim N(\mu = 70, \sigma = 12/\sqrt{9} = 4)$. Therefore: $D = \bar{X}_1 - \bar{X}_2 \sim N(\mu = 75 - 70 = 5, \sigma = \sqrt{2^2 + 4^2} = \sqrt{20})$.

(a) 0.5885

(b) 0.2564